Queen Mary University of London
School of Engineering and Materials Science

Module:	MAT602 & MTRM011 First Sit	**Title:**	Materials Selection in Design
First examiner:	Prof James Busfield	**Second examiner:**	Raza Shah

Qu. No.	Worked Solution	Marks
Section A - MCQ	1. Which of the following materials groups contains materials that are the most difficult to recycle? a. Glasses b. Thermoplastic polymers c. Polymer composites x d. Metals	25
	2. What are the typical units for the toughness of a material, which is also often assigned the symbol G_c? a. J/m^3 b. J/m^2 x c. J/m d. J/m$^{0.5}$	
	3. The attractive forces between atoms in solids help determine several different materials behaviours. Which of the following statements about the relationships between properties can be seen to be true as a consequence of this? a. The thermal expansion coefficient is broadly related to the strength of a material. b. The Young's modulus is broadly determined by the melting temperature of a material. x c. The toughness of a material is broadly determined by the Young's modulus of the material. d. The electrical conductivity of a material is determined by its atomic number.	
	4. The Wiedemann-Franz law is best summarised by which of the following statements? a. The elastic modulus of materials are linearly related to their yield strength. b. The strength of materials are related to their density. c. The electrical and thermal conductivities of materials are linearly related. x d. The toughness and thermal conductivity of materials are linearly related.	
	5. Which of the following materials index values should be maximised to identify the best materials for a lightweight but stiff beam? E is the Youngs Moudlus and ρ is the material density. a. $E^{1/3}/\rho$ b. E/ρ c. E^2/ρ d. $E^{1/2}/\rho$ x	

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	6. Which of the following materials has the lowest density? a. Carbon fibre composites b. Polyethylene x c. Magnesium d. Tungsten carbide	
	7. Which of the following materials has the highest modulus? a. Diamond x b. Titanium alloys c. Steel alloys d. Silicon carbides	
	8. Which of the following does a designer aim to maximise when designing an efficient spring? a. Mass of the spring b. Strength of the spring c. Modulus of the spring d. Energy stored per unit mass of the spring x	
	9. The crystalline structures that form in metals and ceramics are not perfect; they contain defects that are known as? a. Relocations b. Dislocations x c. Linear defects d. Polar defects	
	10. Which of the following statements is true about hot working of metals when compared to cold working? a. Hot working increases the amount of further plastic deformation that is possible. x b. Recrystallisation during hot working locks in the distribution of defects. c. Hot working produces a material with a higher hardness than cold working. d. Recrystallisation is faster at lower temperatures.	
	11. Which of the following statements is the most correct when considering how to ensure that the fibres within a fibre reinforced polymer composites are effectively wet out? a. The resin should have a low viscosity. x b. The resin should have a low thermal conductivity. c. The resin should have a low specific heat capacity. d. The resin should have a low thermal expansion coefficient.	

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	12. Fillers such as talc or calcium carbonate are frequently added to polymer composite materials. Which of the following statements most accurately reflects an important motivation for including these types of fillers into the compound? a. To increase the toughness of the composite material. b. To increase the thermal shrinkage from the mould. c. To reduce the density of the composite material. d. To increase the modulus of the composite material. x	
	13. The largest volume usage for polymer materials globally is in which of the following industries? a. Packaging x b. Automotive c. Electrical goods d. Agriculture	
	14. In a 2012 European Union study which of the following is the most common end of life pathway for polymers? a. Recycling b. Energy recovery c. Landfill disposal x d. Composting	
	15. In excess of 50% of the weight of the Boeing 787, Dreamliner is made up from which of the following materials types? a. Aluminium alloys b. Titanium alloys c. Steel alloys d. Polymer composites x	
	16. Optimum packing for rigid spheres of identical size results in a void volume fraction of how much? a. 30% x b. 23% c. 16% d. 10%	
	17. Which of the following most accurately describes the vapour phase sintering processes? a. The transport mechanism is viscous flow and solution precipitation. b. The transport mechanism is evaporation and condensation. x c. The transport mechanism is viscous flow and diffusion. d. The transport mechanism is diffusion.	

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Qu. No.	Worked Solution	Marks
	18. Which of the following ceramic materials can be used as a heat sink for a transistor device where rapid heat dissipation and low electrical losses are important? a. Aluminium oxide, Al_2O_3 b. Barium titanate, $BaTiO_3$ c. Molybdenum disilicide, $MoSi_2$ d. Aluminium nitride, AlN x	
	19. Which of the following examples can be analysed using a transient (sometimes known as an explicit dynamics) finite element analysis code? a. The heat build-up in a missile due to air friction. b. The frequency analysis of a bridge subjected to wind buffeting. c. The crash worthiness of a car chassis. x d. Stress analysis of a prosthetic hip joint.	
	20. One additive manufacturing process uses a laser beam guided by a mirror to trace a single layer onto the surface of a vat of liquid polymer to cure the polymer. The cured material is then lowered a small distance (typically about 0.1mm) below the remaining liquid, recoating it with more resin and the process is repeated. This process is commonly known as? a. Selective laser sintering b. Stereolithography x c. Laminated object manufacturing d. Fused deposition modelling	
	21. A value function can be drawn on a trade off plot to determine the optimum solution when conflicting goals exist. For example the cost of a bicycle, *C* can be plotted against the mass of a bike, *M*. These are related using a value function, $V=\alpha M+C$. In this value function, the term α is known as the a. Exchange constant x b. Trade off constant c. Resolution constant d. Mass saving constant	
	22. On a stress-strain graph, the amount of strain energy density required to extend the material under a tensile test corresponds to: a. The gradient of the linear elastic region. b. The strain to break. c. The total area under the stress strain graph to that point. x d. Ultimate tensile strength.	

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Qu. No.	Worked Solution	Marks
	23. When designing a fluid seal to prevent fluids from leaking out from a pump, surface-to-surface contact sealing is most effective between: a. A rough, flat surface and a smooth, uneven surface. b. A rough, uneven surface and a smooth, flat surface. c. A rough, flat surface and a rough, uneven surface. d. A smooth, flat surface and a smooth, flat surface. x	
	24. When designing an outdoor metal pipeline, which of the following metal-to-metal pipeline joint combinations would lead the lowest rate of galvanic corrosion between two interconnecting pipe sections. a. Aluminium to Iron b. Aluminium to Zinc c. Aluminium to Aluminium x d. Aluminium to Copper	
	25. To increase the safe operating pressure of a gas cylinder using the same material and without exceeding the allowable hoop (or circumferential) stress [$\sigma_{allowable}$], design engineers would: a. Reduce the diameter of a cylinder. x b. Increase the diameter of a cylinder. c. Reduce the thickness of the cylinder. d. Increase the length of a cylinder.	

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Section B (first sit) Q 1.

Suggest a suitable design for a glass topped table with a working surface thickness of 0.01m and an area of $0.6m^2$. The design should have just four legs and in addition to the glass top it must also have a lower shelf for magazines. You will have to use a bonded interface between the glass top and the legs.

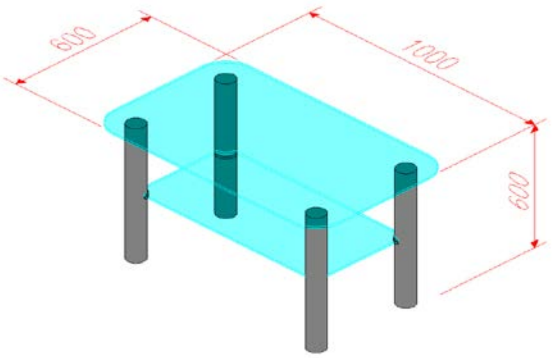
a) Draw a sketch of your overall design including your approximate dimensions. Indicate an estimate for the approximate mass of each component. 5

b) Focusing more closely on the joint interface between the legs and the top surface sketch how the interface will appear and discuss the typical loadings that might result from a range of different abuse loadings. 5

c) Explain giving valid reasons the types of adhesive you would use. 10

d) What are the manufacturing requirements for using that type of adhesive and how might these be achieved during the manufacture of your product. 5

a. Any of a range of designs are possible. But it should have 4 legs – with suggested dimensions and materials.
An identified set of dimensions for the glass shelves, with mass calculated. It must have toughened glass specified.
A sketch of the details for the feet, the end caps and the slots for the mid table. Aluminium is best for light weighting. Extruded and cut tubes are cheap. These can be anodised to give a good finish. The end caps should be a loose push fit and can be machined or cast aluminium. The mid table fixing should also be discussed. Simple slots for the glass are OK. The recesses in the tube and the fixing device must be considered. Table top ~ 20kg; shelf ~15kg; hollow legs <2kg each.



b. Example calculations
Leg assembly (should sketch the end caps and show they are bonded to the inside of the legs)
Adhesive strength $20N/mm^2$
Bond area (circ x height) = $3300mm^2$ approx each leg. Force required to remove cap = 66kN
This is a huge force – They will not come out.

Glass top - Adhesive strength $15N/mm^2$
Typical Bond area = 4 x $3850mm^2$
Force required to remove top = 230kN. This is far greater than the self weight, so you can pick up the table very easily

Other scenarios should include an adult standing in the middle or a child sitting in the middle. (Clearly the bond is much stronger and the glass would break first.) this required not only statics to be solved but also some idea of the bending stress in the glass.
For full credit the student should consider impact forces.

c. The metal end caps are are annaerobiclly bonded inside the aluminium legs as are the mid leg clips recessed slots. This requires a close interference fit to work well. A candidate is Loctite 326. This material is 50% hydroxyethy methacrylate, hydroxylpropyl methacrylate and isobutyl methacrylate, the remainder is acrylic acid and cumene peroxide and cumene. This adhesive has a very short less than 4 minute setting time.

The glass top is UV cured bonding onto the top end caps of the legs. A suitable bonding agent is Loctite 350, which has a very short setting time of less than 20 seconds depending upon the strength of the UV source. This agent is also a methacrylate based material with Isobornyl, hydroxypropyl and lauryl methacrylates making up the bulk of the material. This is coupled with silanes to bond the polymer onto the metal surface and an acrylic acid. The process is also anaerobic.

d. The surfaces must be clean and degreased to make an effective bond. Need controlled quantity of adhesive for all cases, so the operator will have two metered dispensing guns that can mix and store the adhesives carefully. The joint areas need to be filled and this glue needs to be applied carefully as excess must be avoided due to aesthetics. You will see the glass bond throught the service of the table. The table top adhesive requires the use of UV Curing Equipment with a high power lamp for fast cure, which must be guarded to protect the operator. As the table takes a few minutes to build and the legs take a bout 4 minutes to set a carosel is used to make the legs. To give them an appropriate dwell time. The legs are accurately sized prior to assembly with the end caps etc to ensure a good tight interference fit. This makes the bond the most effective.

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Q 2.

The demands continuously increase on automotive bearings resulting from a desire to increase reliability, to reduce size and to help reduce vehicle emissions. The cleanness of the metal is of paramount importance during steelmaking and casting and this especially true when considering the fatigue life for the bearing.

a. Identify three typical grades of steel that are used for bearings and briefly describe each. **5**

b. An industry standard is to use the micro-cleanness Statistics of Extreme Values (SEV) count as defined in ASTM E2283. Explain how the SEV assessment calculates the anticipated maximum non-metallic inclusion size from an examination of a number of photomicrographs of the steel. **5**

c. Many elements have to be carefully controlled during steelmaking to ensure that the final steel is of the correct composition and to ensure that any inclusions are sufficiently small so as not to have a negative impact on the predicted fatigue life. Focussing on Oxygen, Aluminium and Sulphur explain the negative (and potentially any positive) impact that their incorporation might introduce and what measures are taken to control the composition carefully during the steelmaking. **10**

d. Describe how a steel ball bearing is actually made. **5**

a) 100Cr6 (SAE 52100) – a through hardening steel with high fatigue strength and hardness, and good wear resistance.
70SiCrMnMo6-5 – an air hardening steel that has been designed for hardening and quenching to meet range of properties that can be tailored by controlling the heat treatment.
18CrMnMoNi9-5-5 – An air hardening case carburising steel with high toughness and high fatigue strength.

b) Calculate the √area for each inclusion measured;

$\sqrt{area} = \sqrt{(length \times width)}$

Sort the √area values in ascending order; these will become the x-axis values
Calculate the reduced variate Yi for √area; these will become the y-axis values;

$Y_i = -ln[-ln\{i/(n+1)\}]$

Where i = Order which arranged inclusion in the ascending order (e.g.: 1,2,3,...n)
and n = the number of the inclusion measured and calculated

Fit a linear trend line to the data
Using the specified probability value (P) calculate the return period (T_0);

$T_0 = 1/(1-P)$

From the return period, calculate the reduced variate (Y_{max}) required by your specific bearing;

$Y_{max} = -ln\ [-ln\{(T_0-1)/T_0\}]$

Next, determine the reference area (A_{ref}) the extrapolation covers;

$T_0 = A_{ref}/A_0$

A_0 = control area; equivalent to one field observed

Extrapolate √area at the time of Y_{max}

c) Aluminium is a potent deoxidant for steel which is often added for grain refinement. However, a by-product of deoxidisation is the formation of alumina (Al_2O_3) a very harmful non-metallic inclusion species that must be minimised in the final product. Often carbon is used as a primary deoxidant prior to aluminium to reduce alumina formation. Once formed though aluminates are lighter than the liquid steel and given sufficient time will float out of the liquid steel.

Oxygen - Free oxygen will readily oxidise reactive elements such as aluminium and hence must be reduced during steelmaking. Often a maximum content is specified by bearing manufacturers as it is indicative of steel cleanliness. The reoxidation due to environmental ingress must be avoided. The use of refractory hollow ware and engineered powders protect the liquid steel from the environment to avoid this.

Sulphur - Sulphur is controlled in steelmaking by the addition of manganese to form manganese sulphides. Whilst more deformable than aluminates, manganese sulphides are still detrimental to fatigue life, and as aluminate contents are reduced in the steel, these become the dominant fatigue initiation species. To offset this, maximum sulphur contents is also stipulated for bearing steels.

d) A thick piece of wire is fed into a machine where it is cut down by slicing sections down. The machine then slams two hemisphere cavities (a bit like a mould) into the piece of wire making a ball shape. As a result of this process, the ball will have a ring of excess metal around it, called flash, which needs to be removed. The ball also needs further polishing to make it perfectly round and smooth. The ball with the flash is then fed into another machine which rolls the ball around between two rill plates. Rill plates are two hard plates of steel which wears away the flash and smoothens out the surface. The ball is then heated to harden it after which it undergoes a grinding process until it has the final, very accurate measurements (with a required tolerance as small as 1μm). The last process is called lapping, which requires a similar machine that exerts less pressure combined with a polishing paste to give the balls their perfect shiny finish without further reducing their size.

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Q 3.

Ford is redesigning their UK manufactured diesel engines. The sump is the main reservoir of the engines oil at the bottom of the engine. One of the key requirements for the new engine is for the sump to hold an additional 2.5 litres of oil to increase the service intervals for the vehicle.

a. Describe the top ten key criteria to consider when changing the design of the sump. 10

b. The current choice of materials for the sump is to either use the traditional steel used in the current design or to use a new polymeric material. Which material was chosen for the redesign? You should outline all the specific advantages and disadvantages for this material choice. 10

c. Describe using a suitable sketch the process used to manufacture the new sump. 5

The answer needs to be more than a simple bullet list for the full marks

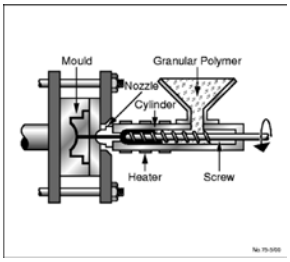
a) The basic function of the plastic oil pan is to store and seal the engine oil from the environment. It must therefore have:

1. It must withstand high temperatures as well as chemical ageing in oil, fuels and exhaust gases and it must have excellent resistance to hydrolysis, especially in glycol.
2. Good stiffness properties at 200C.
3. Resistance to E85 (Blended biofuels) - Understand the effects on both the tensile strength and modulus.
4. Resistance to sulphuric acid, hydrochloric acid and nitric acid (pH 1,5) during heat ageing at different temperatures. This is required to simulate the conditions inside air ducts when exhaust gases are passed back from the engine. Tensile and impact test performance of the material should be considered.
5. Fatigue test - No crack, deformation or degradation allowed at any place of the oil pan. Pressure test after the fatigue test - the test would comprise of. no leakage at 5 PSI (0.34 bar) at the end of 20 million cycles.
6. Lightweight – To provide CO_2 reduction benefits
7. Quality – The product should meet the lifetime requirements for the customer without degrading and causing a breakdown.
8. Cost – It is essential to pick a material that will perform all the required functionality of the part, however the best cost solution should be identified.
9. NVH (Acoustic performance) – This is important for the customers quality perception of the product. The speed that the part will warm up to operating temperature is important.
10. Stone impact test – The part will have to be resistant to an impact test.
11. Strong enough to support misuse such as the drop weight of the engine when the engine is resting on the ground

b) Ford used a short chopped glass filled (35%) polyamide PA6 for the actual part.

1. The reasons being that an increased volume metal sump required the design to be at least a two part (the top part being cast and the lower part being a metal pressing) that would require additional room temperature vulcanised rubber gasket.
2. The plastic sump was nearly 1kg lighter.
3. The plastic sump was moulded in mostly a single moulding with a vibration welded panel on each side,
4. Steel sumps are widely used in practice. Plastic ones are not widely used so far.
5. The plastic sump was easier to assemble and had a simple rubber gasket seal to the engine.
6. Competitors have used plastic sumps in commercial vehicles in the past but Ford only has very limited experience of their use.
7. Plastic offers a simpler design and lower cost tooling.
8. Plastic allows some additional design features like the internal honeycomb shape to be introduced which further reduces weight and improves stone chip protection.
9. The plastic component requires much more careful consideration of the heat shielding especially around the region of the exhaust gases.
10. The heat dissipation in the metal is also better than in the plastic.

c) The plastic sump is manufactured in one main part and 2 smaller flat parts by injection moulding. The three parts are bonded together using vibration welding. A perfect answer should have a sketch of the equipment and describe how the plastic is heated up in the heated barrel. Pumped forward during injection. Cooled down in the mould, ejected when the mould is opened.



Irritating that so many answers talked about 1520kg sumps.

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Section C 1. — 5

An oar can be considered to be a strong, light, stiff beam that is loaded in bending. Its basic requirement is that it must carry the bending moment exerted by the oarsman without breaking. Its stiffness is measured by hanging a weight of 10kg on the oar, 2.05m from the collar and the deflection at the point of loading is measured, as shown below.

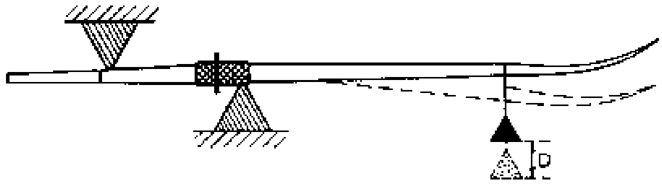

Figure 1.

Assuming that the oar is of uniform cross sectional area, *a*, has a length *L*, and a second moment of area, *I*. The denisty of the material it is made from is ρ. What is the mass, *m* of the oar?

Mass, $m = aL\rho$……………………………………………………………(1)

Assuming that the deflection of a simply supported beam, *D* when it is subjected to a load *F* is

$$D = \frac{FL^3}{C_1 EI}$$

calculate the stiffness of the beam. *S*.

$S = \frac{F}{D} = \frac{C_1 EI}{L^3}$……………………………………………………………………..(1)

For a circular cross section $I = \frac{a^2}{3\pi}$. By eliminating the cross section area, *a* derive an equation for the mass of the oar.

$S = \frac{C_1 E a^2}{3\pi L^3}, a = \sqrt{\frac{3\pi S L^3}{C_1 E}}$…………………………………………………………………...

$m = \left(\frac{3\pi S}{C_1}\right)^{\frac{1}{2}} (L)^{5/2} \left(\frac{\rho}{E^{1/2}}\right)$…………………………………………………………...(1)

Separate this equation up into functional, geometric and materials components.

$m = \left(\frac{3\pi S}{C_1}\right)^{\frac{1}{2}} (L)^{5/2} \left(\frac{\rho}{E^{1/2}}\right)$…………………………………………………………...(1)

Identify what the materials index is that optimises the design of the oar

$M_1(\text{minimum}) = \left(\frac{\rho}{E^{1/2}}\right)$…………………………………………………………...(1)

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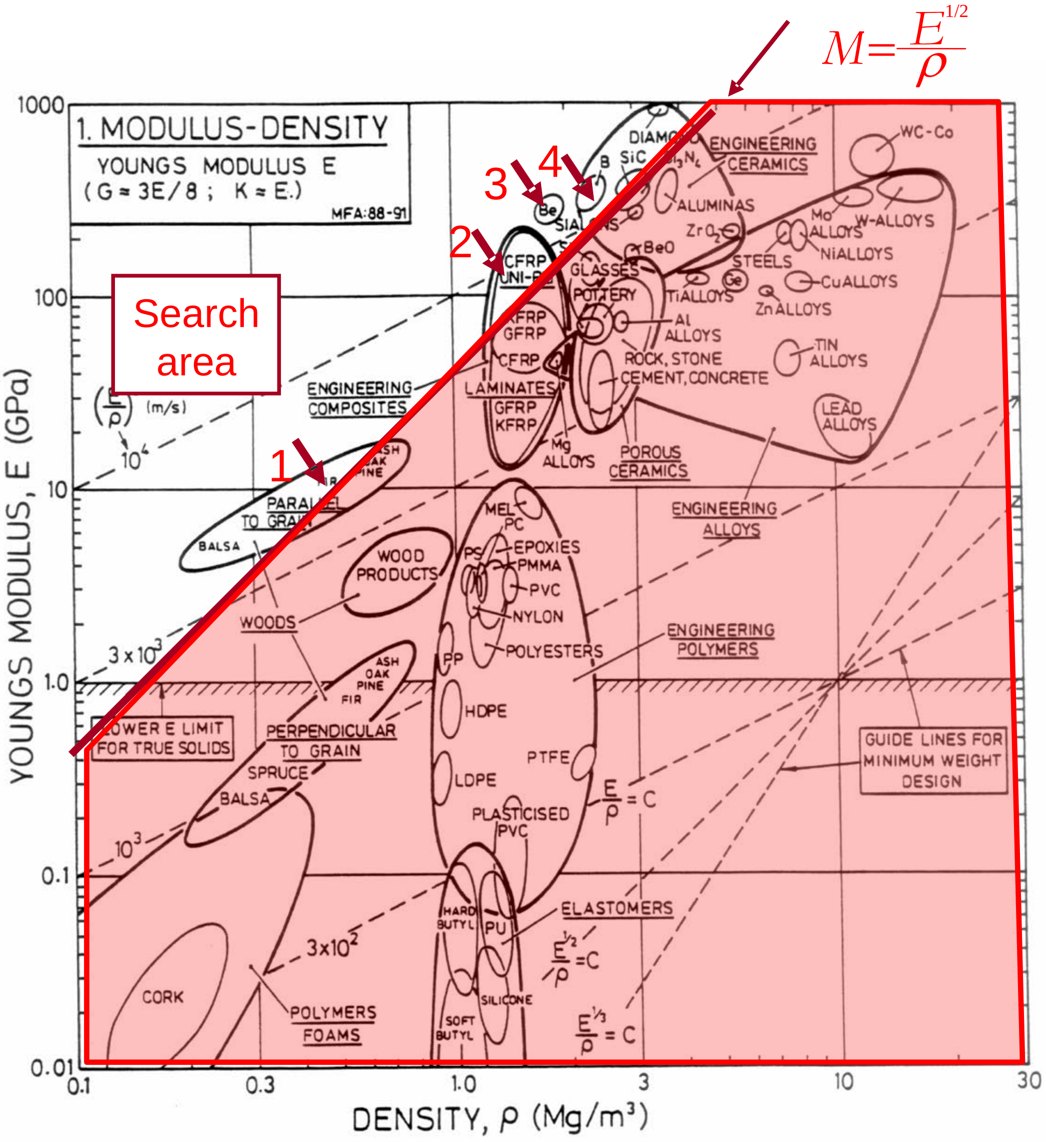
2	Using the Ashby diagram below, highlight the most suitable materials from a range of materials classes that can be used as a minimum mass for the oar described in question 1. 	3

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3

Tabulate below the top five materials for use as an oar from the Ashby diagram in Question 2. Calculate the materials' index for that material and describe the suitability of the materials selected for use as an oar.

4

Selected Material	Index / Units	Could also be $GPa^{0.5}/Mgm^{3}$	Comments on suitability
Wood	$150kg/(m^{3}GPa^{0.5})$	7	Cheap, light, but variable
CFRP	$100kg/(m^{3}GPa^{0.5})$	10	The best choice, more control of
Beryllium	$120kg/(m^{3}GPa^{0.5})$	8	Cost (and toxicity) rule it out
Ceramics	$200kg/(m^{3}GPa^{0.5})$	5	But fracture toughness inadequat
Diamond	$100kg/(m^{3}GPa^{0.5})$	10	Ridiculous choice

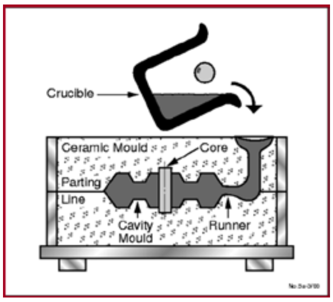
4

Describe using a sketch the most suitable method to manufacture a typical steel engine block for a typical passenger car. Describe the essential elements of the process.

6

Process: Sand casting

Sketch:



Description:
A pattern (model) of the engine block
This is used to make an impression in an expendable sand mould.
The molten steel is poured into the cast and allowed to cool.
The mould is destroyed to remove the part after it has solidified.
The sand can be each use but the pattern to make the mould cavity is re-used.

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5

A 100g polymer component can be manufactured by either machining from an extruded bar or by injection moulding from extruded pellets. At this stage it is not known from your marketing department the precise production quantities but you have been asked to calculate the cost per part of manufacturing 1,000, 10,000 and 100,000 by both processes ignoring the capital and labour costs whilst assuming the following information.

7

Polymer material cost - £5.00 /kg
The waste material created per part by machining - 150g.
The amount of waste material created per part by injection moulding – 5 g.
The investment required to machine these parts is - £1,000.
The injection moulding tooling costs dedicated for moulding this part are - £15,000.

(Show your workings here.)

number	Machining	Moulding
1	1001.25	15000.53
10	1012.5	15005.25
100	1125	15052.5
1000	2250	15525
10000	13500	20250
100000	126000	67500
1000000	1251000	540000

What process is the most appropriate for manufacturing smaller production volumes?

Machining is the best solution at small volumes.